Comment on Garrett, Ohrn, and Suárez Serrato (2020)

Tax Policy and Local Labor Market Behavior, AER Insights

Garrett, Ohrn, and Suárez Serrato (2020) examine the impact of federal bonus depreciation policy on local labor market outcomes using a difference-in-differences framework. By leveraging variation in county-level exposure to investment incentives—driven by cross-sectional differences in industry capital intensity, the authors estimate the employment and earnings effects of this federal tax reform over time. The paper offers valuable empirical evidence that bonus depreciation increases employment in high-exposure counties, but does not raise earnings-per-worker, implying a nuanced and potentially incomplete link between capital policy and labor market welfare.

The empirical strategy is well-designed, with strong identification logic, dynamic modeling, and a range of robustness checks. Nonetheless, the paper presents several methodological and conceptual limitations, particularly with regard to SUTVA assumptions, heterogeneous treatment effects, and the interpretation of temporal dynamics.

Strengths of the Empirical Approach

- Credible Quasi-Experimental Design:

The identification strategy relies on exogenous federal policy changes and cross-sectional variation in exposure, determined by pre-existing industry capital structures. This design minimizes concerns of endogeneity.

- Difference-in-Differences with Dynamic Extensions:

The authors estimate both average treatment effects (Equation 1) and a year-by-year event study (Equation 2). The latter allows for inspection of parallel trends and identification of dynamic responses across different policy regimes (2002 and 2008).

- Robust Fixed Effects Framework:

The inclusion of county, state-by-year, and industry-by-year fixed effects addresses a wide range of potential confounders, ensuring internal validity.

- Policy Relevance and Cost Assessment:

By estimating a cost-per-job of $20,000–$50,000, the paper connects empirical results with fiscal evaluation, benchmarking bonus depreciation against alternative labor market interventions.

Limitations and Areas for Improvement

1. SUTVA and Interference: Although treatment is not assigned at the individual level, the SUTVA assumption may still be violated. County-level outcomes may be influenced by spillovers due to commuting, migration, or firm reallocation across regions. Additionally, there is potential heterogeneity in how firms utilize bonus depreciation—some may invest in labor-intensive capital, while others may substitute toward automation—introducing variation in treatment implementation.

2. Lack of Heterogeneity Analysis: While the paper distinguishes between the 2002 and 2008 policy windows, it does not investigate heterogeneous effects across all industries, worker types, or regional characteristics. This limits insight into the distributional consequences of bonus depreciation. For example, employment gains may be concentrated in low-skill or temporary positions, which could explain the absence of earnings growth.

3. Over-Specification and Power Loss in Dynamic Models: Equation (2) provides a flexible event-study framework but includes a large number of year-specific interaction terms. This may result in overfitting, reduced precision, and difficulty in interpretation. While the use of this specification is helpful for pre-trend testing, its reliability for drawing inference on long-run effects is constrained by potential multicollinearity and small effect sizes.

4. Two-Way Fixed Effects with Continuous Treatment: The continuous-treatment TWFE framework may produce biased estimates if treatment effects are dynamic and vary across counties. While the authors use event study plots to mitigate this concern, they do not apply newer approaches (e.g., interaction-weighted estimators or decomposition methods) that account for TWFE's limitations under staggered treatment and heterogeneous effects.

5. No analysis of investment composition: The paper does not explore whether employment gains came from labor-intensive vs. automation-heavy sectors. If a large share of investment went to capital-deep industries, the policy may have created fewer jobs per dollar . This limits interpretation of the employment effect's quality and inclusiveness.

6. First, our estimates are based on cross-sectional variation that does not account for general equilibrium effects or other major effects of the policy. Second, our estimates do not account for migration induced by the policy (given in paper)

Interpretation of Event Study Results

The event study coefficients suggest that counties with high exposure to bonus depreciation experienced stronger employment growth during the 2002–2004 policy window. These effects diminish over time and appear weak or absent during the 2008–2010 period. Importantly, the flat pre-treatment coefficients (1997–2001) support the parallel trends assumption. Nonetheless, the lack of persistent effects and the absence of wage growth suggest that the employment benefits may be transitory or reflect low-quality job creation.

Evaluation

Overall, the paper is a well-executed and policy-relevant empirical contribution. It demonstrates that capital tax policy can have real effects on employment at the local level. However, the disconnect between employment gains and earnings growth raises important distributional concerns. Future research could benefit from richer data on worker-level outcomes, as well as structural modeling to explore long-run equilibrium effects.

In sum, Garrett, Ohrn, and Suárez Serrato (2020) provide compelling evidence that capital tax incentives can influence labor markets in the short run. Yet, the effectiveness and equity of such policies remain contingent on their design, implementation, and the types of jobs they generate.